1. Subprogram

- A **subprogram** is a small, separate block of code that performs a specific task.
- It is written once and can be **used many times** from different parts of a program.

Subprograms help to:

- break large programs into smaller parts
- reuse code
- make programs easy to read
- simplify testing and debugging
- avoid repetition
- improve program structure

Common Names for Subprograms

In different languages, subprograms are called:

- **functions**
- **procedures**
- **methods**

Main Types of Subprograms

There are two basic kinds:

1. **Function**
   - returns a value
   - usually used inside expressions

   Example:

   sum = add(a, b)

2. **Procedure**
   - performs actions
   - may or may not return a value

   Example:

   printResult(total)

Parts of a Subprogram

A subprogram has:

1. **Header** – name and parameters
2. **Body** – statements to execute
3. **Return value** (for functions)

Parameters and Arguments

- **Parameters** → variables in the subprogram definition
- **Arguments** → actual values passed when calling

Example:

```
function square(x)    // Definition
    return x*xy


square(5)   // Call
```

Here, x is a parameter and 5 is an argument.

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7) Local and Global Variables

- **Local variables** → declared inside the subprogram, used within that block of code.
- **Global variables** → declared outside, accessible everywhere in the program.

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8) Advantages of Subprograms

- improve modularity
- reduce duplication
- make maintenance easy
- improve readability
- simplify large systems